

Lab 3 Report - Kubernetes and Minikube

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Experiment	Lab 3: Kubernetes tutorials with Minikube
Execution	GitHub Actions Ubuntu runner with Docker, minikube, kubectl and istioctl
Evidence repository	https://github.com/whatokk/cloud-labs-03-04-kubernetes-istio
Successful Actions run	https://github.com/whatokk/cloud-labs-03-04-kubernetes-istio/actions/runs/27429016540
Evidence log	D:\论文\期末作业\计算机\Lab3_Lab4_Kubernetes_Istio\evidence\lab3-lab4-evidence\lab3-lab4-run.log

Objective

Follow the Kubernetes Hello Minikube and Kubernetes Basics tutorials. The experiment starts a minikube cluster, deploys applications, explores pods and logs, exposes a NodePort service, scales the deployment, performs a rolling update, and rolls the deployment back.

Major Steps and Results

1. Started minikube on the Docker driver and verified cluster-info and node status. 2. Created hello-minikube and exposed it through NodePort; curl returned the echo server response. 3. Deployed kubernetes-bootcamp, inspected pod details, logs, and environment variables. 4. Exposed bootcamp as NodePort and verified HTTP access. 5. Scaled bootcamp to 4 replicas and observed requests served by multiple pods. 6. Updated the deployment image to nginx:1.27-alpine, confirmed successful rollout, then rolled back to gcr.io/google-samples/kubernetes-bootcamp:v1.

Screenshot 1. Hello Minikube deployment, NodePort exposure, and curl verification.

```
Lab 3 - Hello Minikube
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## kubectl create deployment hello-minikube --image=registry.k8s.io/echoserver:1.10
deployment.apps/hello-minikube created

## kubectl expose deployment hello-minikube --type=NodePort --port=8080
service/hello-minikube exposed

## kubectl wait --for=condition=available --timeout=180s deployment/hello-minikube
deployment.apps/hello-minikube condition met
hello-minikube URL: http://192.168.49.2:32338

Hostname: hello-minikube-6fcb694474-xrcxt

Pod Information:
❏no pod information available-

Server values:
❏server_version=nginx: 1.13.3 - lua: 10008

Request Information:
❏client_address=10.244.0.1
❏method=GET
❏real path=/
❏query=
❏request_version=1.1
❏request_scheme=http
❏request_uri=http://192.168.49.2:8080/

Request Headers:
❏accept=/*/*
❏host=192.168.49.2:32338
❏user-agent=curl/8.5.0

Request Body:
❏no body in request-

## kubectl get deployments,pods,services -o wide
NAME                                READY  UP-TO-DATE  AVAILABLE  AGE  CONTAINERS  IMAGES
SELECTOR
deployment.apps/hello-minikube      1/1    1           1           8s   echoserver
registry.k8s.io/echoserver:1.10    app=hello-minikube

NAME                                READY  STATUS  RESTARTS  AGE  IP            NODE  NOMINATED
NODE  READINESS GATES
pod/hello-minikube-6fcb694474-xrcxt  1/1    Running  0          3s   10.244.0.3    minikube  <none>
<none>

NAME                                TYPE           CLUSTER-IP      EXTERNAL-IP  PORT(S)          AGE  SELECTOR
...
```

Screenshot 2. Bootcamp scaling, rolling update, and rollback evidence.

```
Lab 3 - Scale
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## kubectl scale deployments/kubernetes-bootcamp --replicas=4
deployment.apps/kubernetes-bootcamp scaled

## kubectl wait --for=condition=available --timeout=180s deployment/kubernetes-bootcamp
deployment.apps/kubernetes-bootcamp condition met

## kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
hello-minikube      1/1     1            1           20s
kubernetes-bootcamp 4/4     4            4           12s

## kubectl get pods -o wide
NAME                READY   STATUS    RESTARTS   AGE   IP            NODE           NOMINATED
NODE   READINESS GATES
hello-minikube-6fcb694474-xrcxt 1/1     Running   0           15s   10.244.0.3    minikube      <none>
<none>
kubernetes-bootcamp-67fbdd6b79-5dcwx 1/1     Running   0           1s    10.244.0.5    minikube      <none>
<none>
kubernetes-bootcamp-67fbdd6b79-jsbth 1/1     Running   0           12s   10.244.0.4    minikube      <none>
<none>
kubernetes-bootcamp-67fbdd6b79-nhtjt 1/1     Running   0           1s    10.244.0.7    minikube      <none>
<none>
kubernetes-bootcamp-67fbdd6b79-zxh5s 1/1     Running   0           1s    10.244.0.6    minikube      <none>
<none>
Hello Kubernetes bootcamp! | Running on: kubernetes-bootcamp-67fbdd6b79-jsbth | v=1
Hello Kubernetes bootcamp! | Running on: kubernetes-bootcamp-67fbdd6b79-5dcwx | v=1
Hello Kubernetes bootcamp! | Running on: kubernetes-bootcamp-67fbdd6b79-jsbth | v=1
Hello Kubernetes bootcamp! | Running on: kubernetes-bootcamp-67fbdd6b79-5dcwx | v=1

=====
Lab 3 - Update
=====

## kubectl scale deployments/kubernetes-bootcamp --replicas=2
deployment.apps/kubernetes-bootcamp scaled

## kubectl wait --for=condition=available --timeout=180s deployment/kubernetes-bootcamp
deployment.apps/kubernetes-bootcamp condition met

## kubectl set image deployments/kubernetes-bootcamp kubernetes-bootcamp=nginx:1.27-alpine
deployment.apps/kubernetes-bootcamp image updated

## kubectl rollout status deployments/kubernetes-bootcamp --timeout=300s
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 1 old replicas are pending termination...
deployment "kubernetes-bootcamp" successfully rolled out

## kubectl describe deployments/kubernetes-bootcamp
Name:                kubernetes-bootcamp
Namespace:           default
CreationTimestamp:    Fri, 12 Jun 2026 16:33:05 +0000
Labels:              app=kubernetes-bootcamp
Annotations:         deployment.kubernetes.io/revision: 2
Selector:             app=kubernetes-bootcamp
Replicas:            2 desired | 2 updated | 2 total | 2 available | 0 unavailable
StrategyType:        RollingUpdate
MinReadySeconds:     0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=kubernetes-bootcamp
  Containers:
    kubernetes-bootcamp:
      Image:      nginx:1.27-alpine
      Port:       <none>
      Host Port:  <none>
      Environment: <none>
      Mounts:      <none>
      Volumes:      <none>
      Node-Selectors: <none>
      Tolerations:  <none>
Conditions:
  Type           Status  Reason
  ----           -
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  kubernetes-bootcamp-67fbdd6b79 (0/0 replicas created)
NewReplicaSet:   kubernetes-bootcamp-7566cfdcf (2/2 replicas created)
Events:
  Type    Reason              Age   From                  Message
  ----    -
  Normal  ScalingReplicaSet   16s   deployment-controller Scaled up replica set kubernetes-
bootcamp-67fbdd6b79 from 0 to 1
  Normal  ScalingReplicaSet   5s    deployment-controller Scaled up replica set kubernetes-
bootcamp-67fbdd6b79 from 1 to 4
...
```

Notes

The original tutorial image `gcr.io/google-samples/kubernetes-bootcamp:v2` did not become ready reliably in the GitHub runner, so the rolling update proof used `nginx:1.27-alpine` as a stable replacement image. The Kubernetes update mechanism was still exercised through `kubectl set image`, `rollout status`, `deployment describe`, and `rollout undo`.

References

Kubernetes official tutorials: [Hello Minikube](#); [Deploy](#), [Explore](#), [Expose](#), [Scale](#), and [Update an application](#).